DEVIXO SOLUTIONS

TASK 04 PROJECT REPORT

GreenCart — Full-Stack E-Commerce System

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Program	BS Information Technology
Project type	E-Commerce System
Technology stack	HTML, CSS, JavaScript, PHP, MySQL
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A responsive learning project demonstrating core e-commerce workflows, authentication, database CRUD, and a role-protected administration area.

1. Introduction

GreenCart is a full-stack e-commerce demonstration developed for Devixo Solutions Task 04. It provides a product catalogue, search and category filters, shopping cart, account registration and login, mock checkout, order history, and an administrator dashboard for product management.

2. Objectives

• Build a responsive and professional storefront.
• Connect a PHP backend to a MySQL database.
• Implement user registration, login, logout, and protected pages.
• Demonstrate Create, Read, Update, and Delete (CRUD) operations.
• Add search, category filtering, cart, order placement, and order history.
• Apply basic security practices such as password hashing, prepared statements, server-side validation, and CSRF tokens.

3. Technology Stack

Layer	Technology	Purpose
Frontend	HTML5, CSS3	Responsive layout and styling
Client-side	JavaScript	Mobile navigation, table search, confirmation
Backend	PHP	Sessions, authentication, business logic
Database	MySQL	Users, products, orders and order items
Local server	XAMPP	Apache and MySQL for local development

4. System Features

Customer module: account creation, login/logout, product browsing, search, category filter, product detail, cart quantity updates/removal, mock checkout, and order history.

Admin module: dashboard statistics, product creation, editing, deletion, inventory table, and table search. Admin pages are restricted to accounts with the admin role.

Database: relational tables store users, products, orders, and order items. Foreign keys preserve order relationships.

5. Database Design

Table	Main fields	Relationship
users	id, name, email, password_hash, role	One user can place many orders
products	id, name, description, category, price, image_url, stock	Referenced by order_items
orders	id, user_id, total, status, created_at	Belongs to a user
order_items	order_id, product_id, product_name, quantity, unit_price	Belongs to an order; product reference can become NULL

6. Security and Validation

• Passwords are stored with PHP password_hash() and checked using password_verify().
• PDO prepared statements are used for database queries with user input.
• Sessions track logged-in users; session IDs are regenerated at login.
• Admin operations check the account role on the server.
• Forms use CSRF tokens for state-changing requests.
• Output is escaped before being rendered in HTML.
• Product fields and registration details receive server-side validation.

7. Installation and Execution

1. Install XAMPP and start Apache and MySQL.
2. Copy the project folder into C:\\xampp\\htdocs\\Devixo_Task04_Ecommerce.
3. Open phpMyAdmin at http://localhost/phpmyadmin and import database.sql.
4. Check the database credentials in config.php.
5. Open http://localhost/Devixo_Task04_Ecommerce/ in a browser.
6. Register a user. To enable admin functions, change that user's role to admin in the users table, then log out and back in.

8. Testing Checklist

Test case	Expected result
Register with valid details	Account is created and user is signed in
Register with invalid email / short password	Validation message is shown
Login with incorrect password	Login is rejected
Search/filter products	Matching products are displayed
Add/update/remove cart items	Cart contents and totals update
Place demo order	Order and order items are saved; stock is reduced
Open admin page as regular user	Access is denied
Admin adds/edits/deletes product	Catalogue is updated in MySQL

9. Limitations and Future Improvements

This is a local learning demo. Checkout is intentionally a mock payment flow and does not accept card details. Future improvements could include image uploads, email notifications, pagination, order-status management, automated tests, deployment with HTTPS, and a real payment provider integrated under appropriate security controls.

10. Conclusion

GreenCart demonstrates the main concepts required for a full-stack e-commerce application: a responsive interface, PHP backend, MySQL integration, authentication, CRUD operations, dashboard statistics, search, and basic security. The project is intended for local educational use and can be extended as more features are learned.

Prepared for Devixo Solutions Task 04. Please run the application locally and add genuine screenshots of your own running version before final submission if your supervisor requires screenshots.